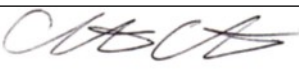


Guideline Title: Pediatric Hypotension

Guideline Number: GUID-3203-PED009

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To recognize the patient with hypotension. To provide and maintain adequate oxygenation, ventilation, and tissue perfusion. To provide adequate blood pressure support and expeditious transport to the nearest appropriate medical facility.
- II. **DEFINITIONS:** Hypotension in children is determined by age & systolic blood pressure (BP) mmHg.
 - A. Term neonates (0 to 28 days): The systolic BP is < 60 mmHg
 - B. Infants (1 to 12 months): Systolic BP is < 70 mmHg.
 - C. Children 1 to 10 years (5th BP percentile): Systolic BP is < 70 mmHg + (age in years x 2).
 - D. Children > 10 years: Systolic BP is < 90 mmHg.
- III. **DEPARTMENT GUIDELINE:**
 - A. Assess airway, breathing, and circulation as per *GUID3203-PED001 – Pediatric General Management*.
 - B. Apply cardiac monitor, monitor pulse, blood pressure, and oximetry. Assess vital signs.
 - C. Assess for clinical signs: tachypnea, tachycardia, altered mental status, and either bounding pulses in hyperemic extremities (warm shock) or peripheral vasoconstriction with weak peripheral pulses, mottling, and delayed capillary refill in cool extremities (cold shock).
 - D. Use a color-coded length-based tape to determine the child's weight (if not known)
 - E. Obtain IV/IO access for the patient's size.
 - F. Treat underlying cause of hypotension: hypovolemia, decreased vascular resistance, heart failure
 - G. Fluid resuscitation
 - i. Administer 0.9% NS or Lactated Ringers 20 ml/kg IV, may be repeated per *GUID3203-PR015 – Intravenous Fluid Therapy*.
 - ii. Most hypovolemic children will require at least 40-60 ml/kg in the first hour and those with sepsis may require up to 80 ml/kg in the first hour.
 - iii. To help deliver fluid rapidly: Place an in-line 3-way stopcock in the IV tubing system. Deliver fluid by using a 30- to 60-mL syringe to push fluids through the stopcock or use a pressure bag (beware of risk of air embolism) or a rapid infusion device.
 - H. If actively hemorrhaging, control bleeding, and consider infusing blood products per *GUID3203-PR002 - Blood Product Administration* or *GUID3203-PR025 – Emergency Blood Product Administration* at 10 ml/kg.
 - I. Utilize fluid warmer when possible per *GUID3203-PR026 Warming of Blood Products & Intravenous Fluids*.

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- J. If hypotension persists after adequate fluid resuscitation, initiate one of the following:
 - i. **Epinephrine infusion at 0.05 -2 mcg/kg/min IV/IO titrate q 5 min to appropriate SBP**
 - ii. **Norepinephrine infusion at 0.03 - 0.5 mcg/kg/min IV/IO titrate q 5 min to appropriate SBP**
 - iii. **Dopamine infusion at 5 - 20 mcg/kg/min IV/IO titrate q 5 min to appropriate SBP**
 - K. If hypotension attributed to heart failure, refer to *GUID3203-PED006 – Cardiogenic Shock*.
 - L. For patients with hypotension due to a dysrhythmia, follow appropriate guideline.
 - M. A child in septic shock with fluid-refractory and epinephrine- or norepinephrine-resistant shock may have adrenal insufficiency. If you suspect adrenal insufficiency, or a patient is at known risk for adrenal insufficiency (i.e., history of steroid use), consider **Hydrocortisone 1-2 mg/kg IV** early from a sending facility. If possible, obtain a baseline cortisol concentration before administration, and obtain expert consultation for any additional evaluation and management.
 - N. Maintain the optimal body temperature.
- IV. DOCUMENTATION:**
- A. EMS Charts
- V. REFERENCES:**
- A. Pediatric Advanced Life Support Manual, 2020. American Heart Association.